String Theory and M-Theory

Lecture 7: Worldsheets, Wick Rotation, and Channel Duality

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Companion notes curated by LazyingArt LLC with Video2Book

Chapter 1

Worldsheets, Wick Rotation, and Channel Duality

Overview. A particle traces a worldline; a string sweeps out a worldsheet. Quantizing either object requires a sum over histories, but the Lorentzian sum is oscillatory and difficult to define. Turning worldsheet time into an imaginary coordinate gives a Euclidean theory governed by the Laplace equation. Its conformal symmetry then turns the slit surface of a four-string interaction into a disc with four boundary insertions, where the symmetry between direct and crossed channels becomes visible.

1.1 From a Worldline to a Sum over Histories

Begin with a free particle. In nonrelativistic mechanics its position is a function $\mathbf{x}(t)$, and, with the mass set to one, its action is

$$S_{\mathrm{NR}}[\mathbf{x}] = \frac{1}{2}\int_{t_1}^{t_2} \dot{\mathbf{x}}^{\,2}\, dt. \tag{1.1}$$

For a relativistic description it is more natural to place time and space in one vector, $X^\mu(\tau)$, parameterized along the trajectory by τ. The corresponding quadratic expression is

$$S_{\mathrm{rel}}[X] = \frac{1}{2}\int_{\tau_1}^{\tau_2} \frac{dX^\mu}{d\tau}\frac{dX_\mu}{d\tau}\, d\tau. \tag{1.2}$$

The upper and lower indices encode the Lorentzian sign difference between time and space. [1]

[1] Editorial clarification: The quadratic relativistic action is being used in a fixed worldline parameterization. The manifestly reparameterization-invariant alternative is proportional to proper length, which is mentioned later in the lecture.

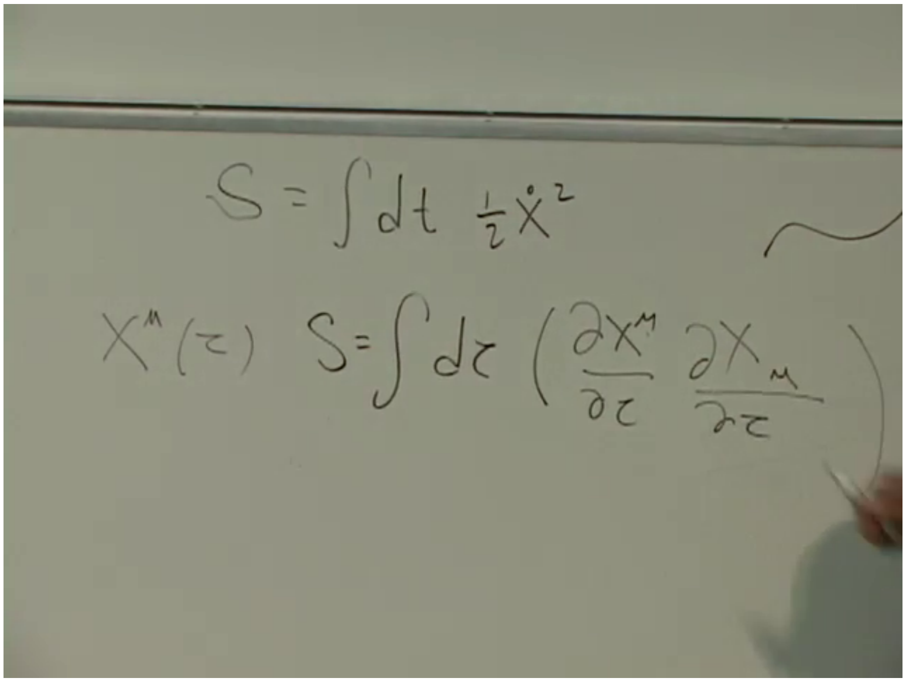

Figure 1.1: The nonrelativistic action and its relativistic quadratic counterpart written together on the board.

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Classically, one fixes the endpoints and varies the path. A free particle then follows a straight worldline with constant velocity. Quantum mechanics asks a different question. It does not select one trajectory between the endpoints; it asks for the amplitude to begin at one point and be detected at the other. Every intervening path contributes a phase,

$$K(2,1) = \int_{X(1)}^{X(2)} \mathcal{D}X \, e^{iS[X]}. \tag{1.3}$$

The functional integral is an integral over an infinite-dimensional space of paths, not the ordinary integral already appearing inside the action.

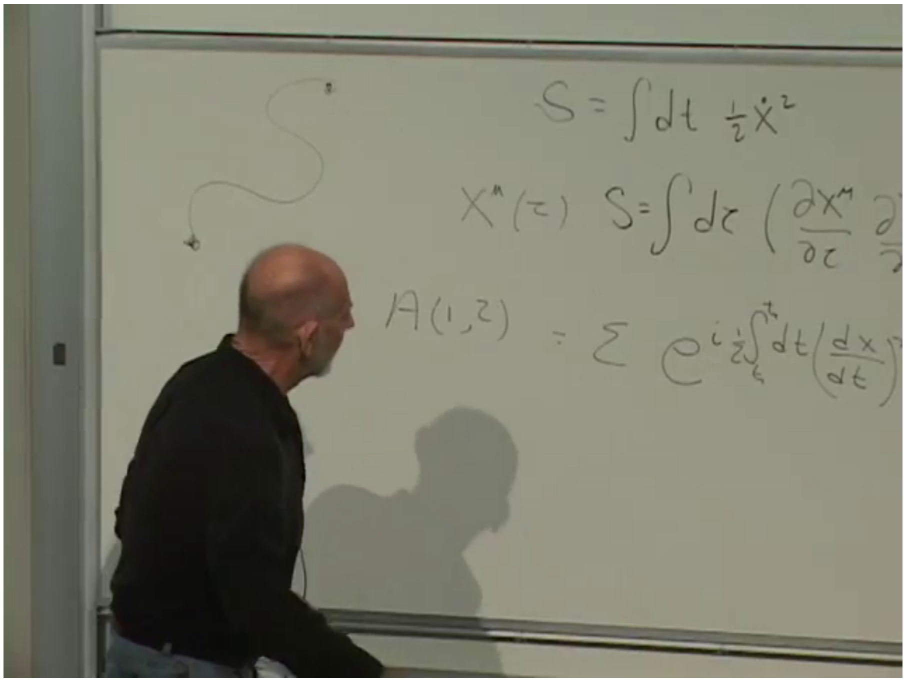

Figure 1.2: A representative path and the phase assigned to it before the sum over all trajectories is taken.

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In relativistic language this point-to-point amplitude is a propagator. For an interaction, several worldline segments meet. Their actions are added, the meeting points are integrated over, and diagrams with the required external particles are summed. Thus the familiar Feynman-diagram expansion is already implicit in the instruction to sum over histories.

The symbol S will denote an action here. It is unrelated both to entropy and to the Mandelstam variable s used near the end of the lecture. The shared notation is traditional rather than illuminating.

1.2 Turning Time Sideways

There is an immediate problem with (1.3). Every factor e^{iS} has magnitude one. A wildly oscillating trajectory and the classical straight line therefore enter with equal absolute weight; only their phases differ. Destructive interference recovers classical behavior, but it does not make the formal integral an ordinary convergent object.

Introduce a new parameter s through

$$\tau = \alpha s, \qquad d\tau = \alpha\, ds, \qquad \frac{dX}{d\tau} = \frac{1}{\alpha}\frac{dX}{ds}. \tag{1.4}$$

Then the quadratic action rescales as

$$\frac{1}{2}\int d\tau \left(\frac{dX}{d\tau}\right)^2 = \frac{1}{2\alpha}\int ds \left(\frac{dX}{ds}\right)^2. \tag{1.5}$$

Choosing the contour corresponding to $\alpha = -i$ changes the oscillatory weight into

$$e^{iS} \longrightarrow \exp\left[-\frac{1}{2}\int ds \left(\frac{dX}{ds}\right)^2\right] = e^{-S_E}. \tag{1.6}$$

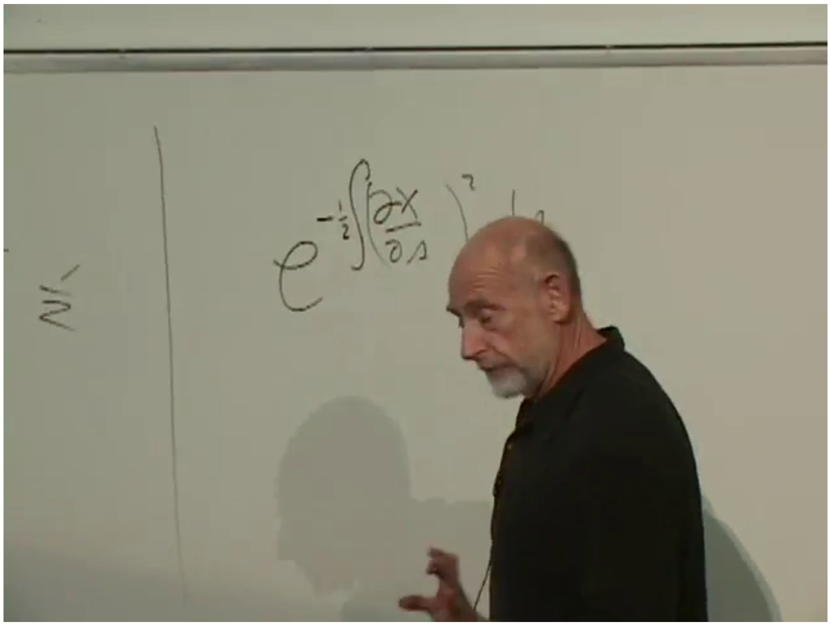

Figure 1.3: The Euclidean particle weight after the imaginary-time substitution; the exponent is real and nonpositive.

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A path that runs far away and returns, or wiggles rapidly without traveling far, has a large positive S_E and is exponentially suppressed. The result is a Wiener-type integral, for which a serious convergence theory exists. The endpoints inherited from real τ lie on an imaginary s-contour. In practice one evaluates the Euclidean object in its convergent domain and analytically continues the answer back to the physical Lorentzian variables. This operation is the Wick rotation: time has been turned sideways for the calculation, not discarded.

1.3 From Worldlines to Worldsheets

A point particle has a worldline. An open string sweeps out a ribbon, and a closed string sweeps out a tube. Coordinates (σ, τ) label points on this two-dimensional worldsheet, while

$$X^\mu = X^\mu(\sigma, \tau) \tag{1.7}$$

gives the spacetime position of the labeled point. Knowing all the functions X^μ specifies the complete history of the string.

The quadratic Lorentzian worldsheet action has the same kinetic-minus- potential structure as an ordinary vibrating string:

$$S_{\rm ws}[X] = \frac{T_s}{2}\int d\tau\, d\sigma\, \left[\partial_\tau X^\mu \partial_\tau X_\mu - \partial_\sigma X^\mu \partial_\sigma X_\mu\right]. \tag{1.8}$$

The overall tension T_s fixes the scale; the lecture suppresses it while concentrating on the derivative structure.

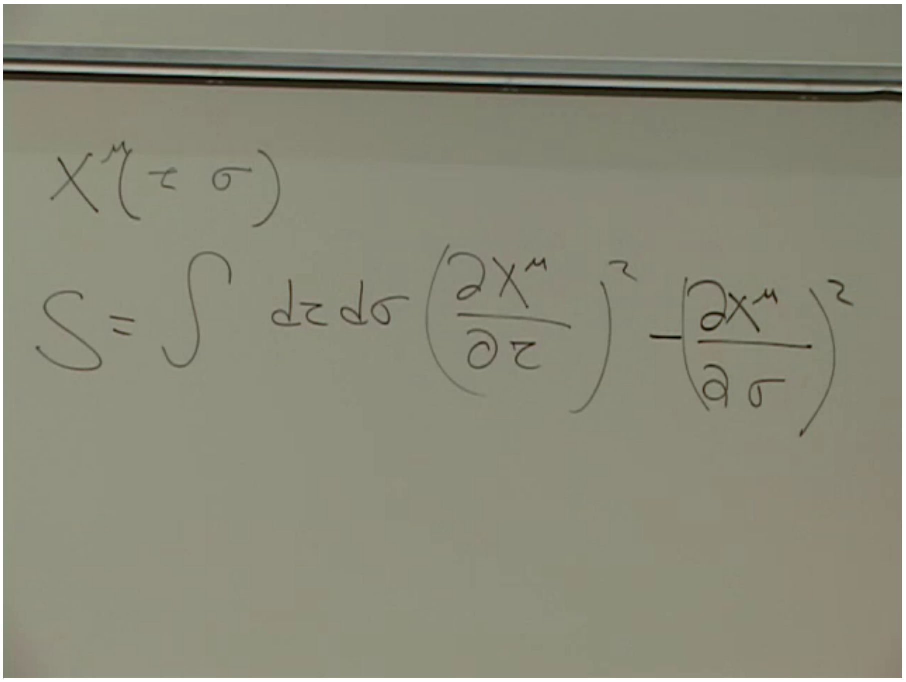

Figure 1.4: The Lorentzian worldsheet action for $X^\mu(\sigma, \tau)$, with its kinetic-minus-potential sign.
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The string amplitude is obtained by exponentiating this action and summing over all surfaces connecting prescribed initial and final string configurations:

$$\mathcal{A} = \int \mathcal{D}X\, e^{iS_{\mathrm{ws}}[X]}. \tag{1.9}$$

The same rule handles two strings becoming two, two becoming three, or any other allowed external configuration. Different worldsheet topologies also belong to the sum. Adding holes is the string counterpart of adding loop orders in a point-particle diagram expansion.

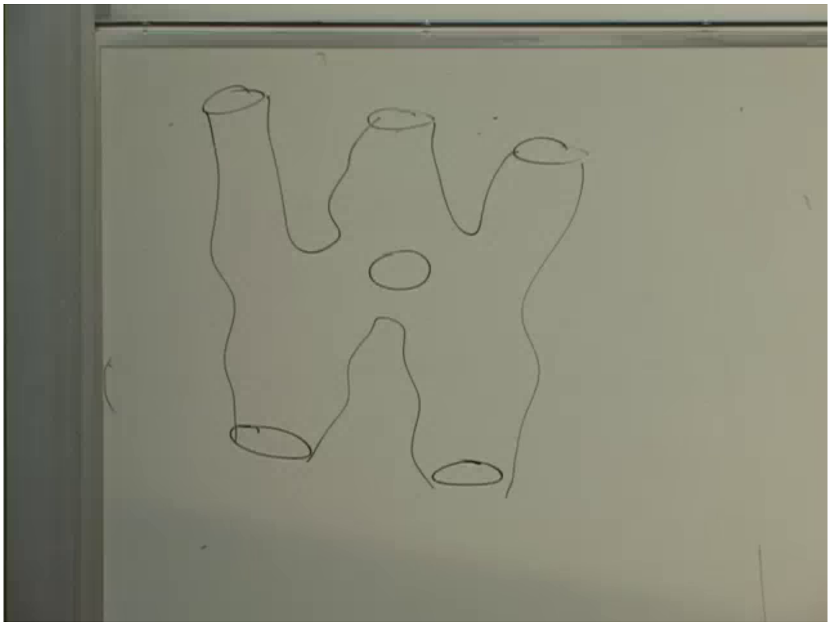

Figure 1.5: A surface connecting several external strings, with a hole representing a different worldsheet topology.

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Question from the audience. For a point electron, does the σ-derivative simply vanish? [a]

Response. A point particle has no σ coordinate in the first place. Nevertheless the pictures are connected: when external states are light compared with the string scale and process energies are low, string amplitudes reproduce ordinary point-particle Feynman diagrams. That limit is more than setting one derivative to zero.

[a] Lecture timestamp: 00:27:56.

The surface integral has the same convergence problem as the path integral. After Wick-rotating worldsheet time, its stable Euclidean form is

$$S_{E,\mathrm{ws}}[X] = \frac{T_s}{2}\int d\tau\, d\sigma\, \left[\partial_\tau X^\mu \partial_\tau X_\mu + \partial_\sigma X^\mu \partial_\sigma X_\mu\right], \qquad \mathcal{A}_E = \int \mathcal{D}X\, e^{-S_{E,\mathrm{ws}}}. \tag{1.10}$$

Both derivatives now enter with the same positive sign. A violently vibrating or greatly stretched surface has large Euclidean action and is suppressed.

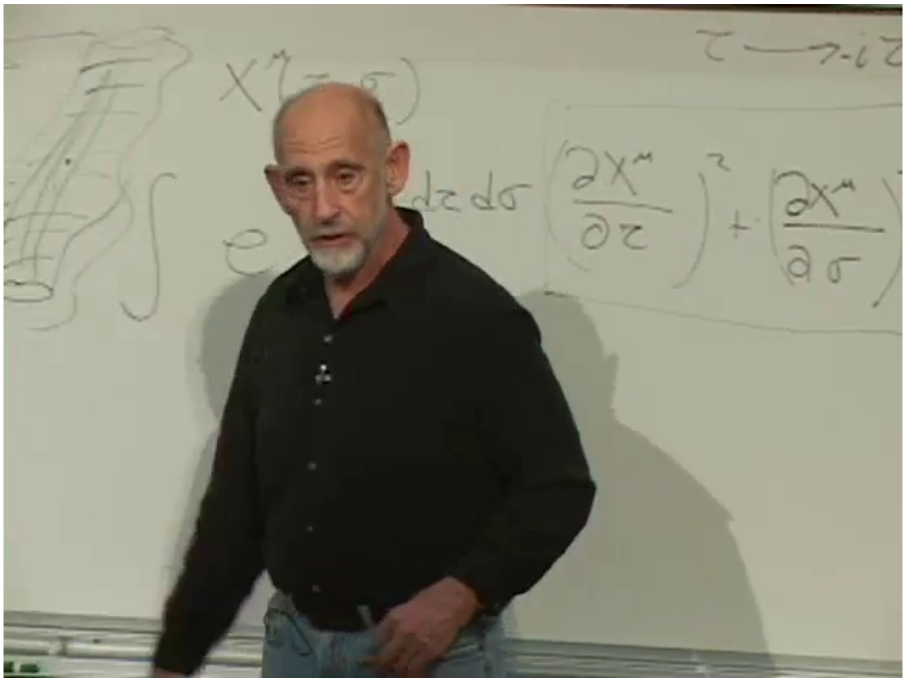

Figure 1.6: The positive Euclidean worldsheet density after τ is rotated to imaginary time.
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There is also a geometrically direct formulation. A relativistic particle may be assigned an action proportional to the proper length of its path; a string may be assigned an action proportional to the spacetime area of its worldsheet. The latter is the Nambu–Goto action. The quadratic worldsheet form used here is conventionally called the Polyakov form and is much easier for the present calculation. [2]

1.4 The Euclidean Equation of Motion

The coordinates σ and τ are only labels on the sheet. The next question is therefore unavoidable: which relabelings leave the theory unchanged? The answer is easiest to see from the equation of motion. Varying (1.10) gives, component by component,

$$\partial_\tau^2 X^\mu + \partial_\sigma^2 X^\mu = 0. \tag{1.11}$$

Before Wick rotation the relative sign is negative and this is a wave equation. In Euclidean signature both signs are positive, so it is the two-dimensional Laplace equation.

Question from the audience. Is this the equation of a vibrating drum? [a]

Response. It is closely related, but a vibrating drum also carries a real time coordinate and therefore obeys a wave equation. The equation here is Euclidean Laplace's equation. Electrostatic potential in a charge-free region supplies a more direct familiar example.

[a] Lecture timestamp: 00:41:50.

[2] Editorial clarification: The lecture includes a humorous discussion of historical naming and also mentions the related Liouville action and Dirichlet boundary conditions. No priority claim is needed for the derivation: only the equivalence of the area and auxiliary-metric formulations is used.

Question from the audience. The action contains first derivatives. Why does its equation contain second derivatives? [a]

Response. Varying a term quadratic in a first derivative and integrating by parts moves one derivative onto that first derivative. It is the field-theory version of $\dot{x}^2$ in an action producing $\ddot{x}$ in the equation of motion.

[a]Lecture timestamp: 00:42:44.

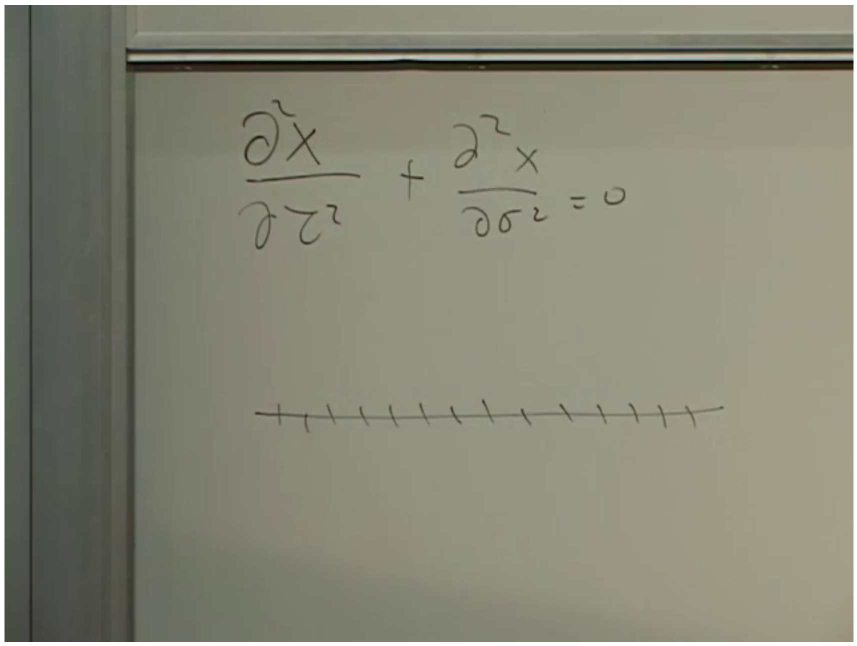

Figure 1.7: The plus-sign Laplace equation and the first discrete line used to interpret its second derivatives.

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To see the geometry, first discretize one coordinate with spacing ϵ. A first derivative is a neighboring difference, while a second derivative is a difference of neighboring differences:

$$\left.\frac{d^2X}{ds^2}\right|_2 \longrightarrow \frac{X(3) - 2X(2) + X(1)}{\epsilon^2}. \tag{1.12}$$

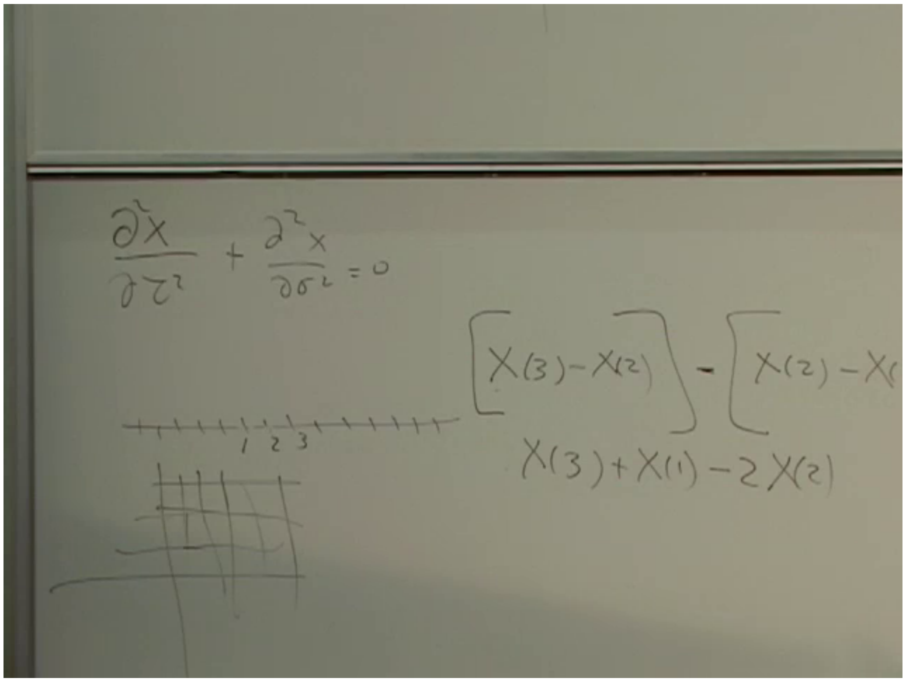

Figure 1.8: The numbered line, small lattice patch, and finite-difference expansion of a second derivative.

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Now discretize the (σ, τ)-plane. Label four nearest neighbors of a central point 5 by $1, 2, 3, 4$. Equation (1.11) becomes

$$X(1) + X(2) + X(3) + X(4) - 4X(5) = 0, \tag{1.13}$$

or

$$X(5) = \frac{1}{4}\big[X(1) + X(2) + X(3) + X(4)\big]. \tag{1.14}$$

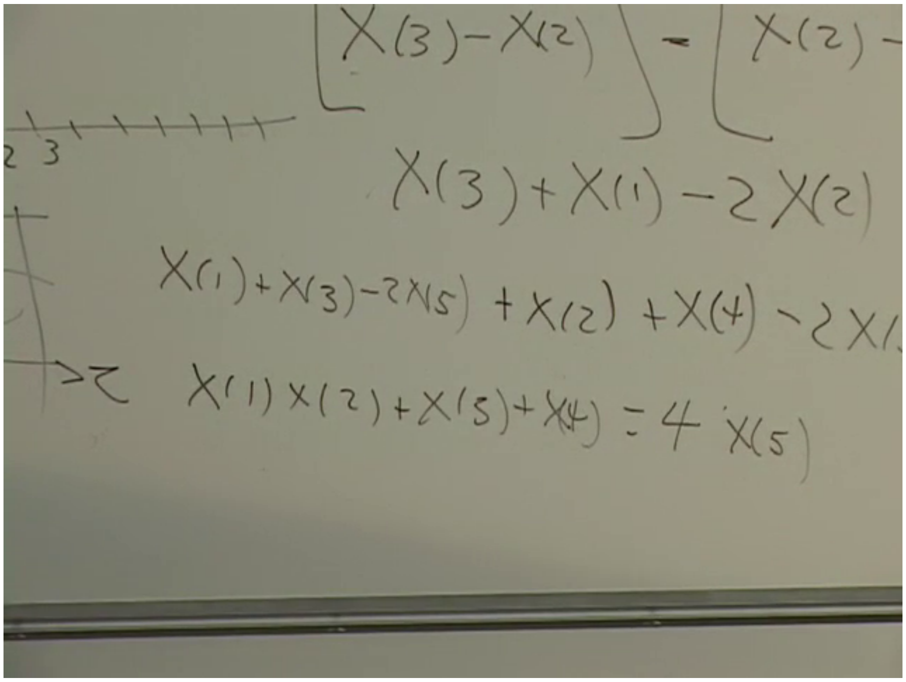

Figure 1.9: The completed two-dimensional finite-difference equation: the central value equals the average of its four neighbors.

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This mean-value property holds for an infinitesimal square in any orientation. It is the local geometric content of the Laplace equation and the bridge to conformal symmetry.

1.5 Conformal Coordinates

Consider a change from (σ, τ) to (σ', τ'). A general deformation sends a small square to an arbitrary quadrilateral. The special transformations relevant here send every infinitesimal square to another infinitesimal square, possibly rotated and rescaled. Equivalently, they preserve the angles at which curves meet. These are conformal transformations.

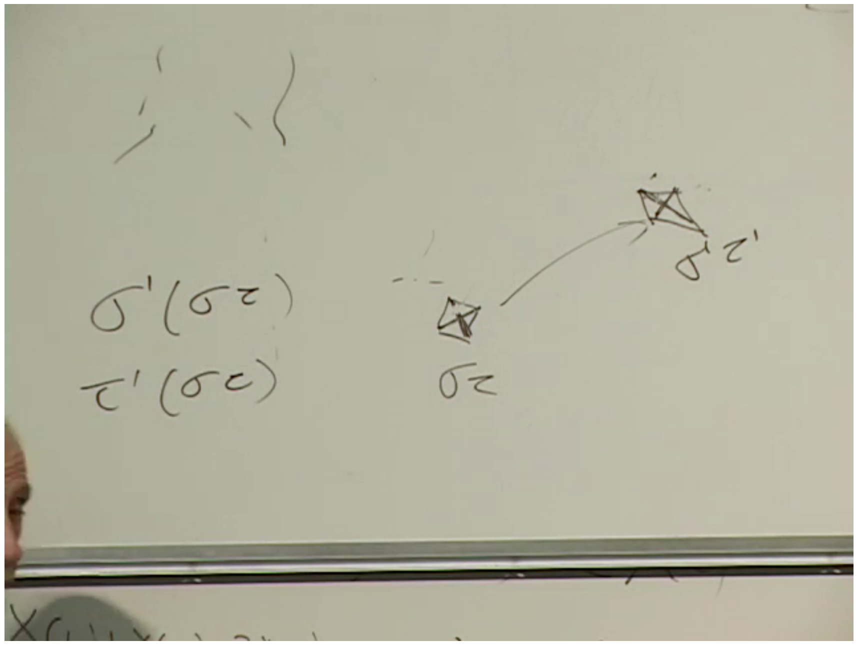

Figure 1.10: A local square mapped to another small figure while its angles are tested for conformal preservation.

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Question from the audience. If angles are preserved, is the transformation basically only a rotation? [a]

Response. Only locally, and even there a common scale change is also allowed. The local rotation and scale factor may vary from point to point, so the coordinate grid can bend, expand in one region, and contract in another while preserving infinitesimal shapes.

[a]Lecture timestamp: 00:53:31.

An angle-preserving map need not preserve size. If size were also fixed everywhere, the possibilities would collapse to the much smaller rigid class of translations and rotations. Conformal maps instead preserve local shape. This is why a map projection can keep the shape of a small island while greatly distorting its area.

Question from the audience. Is a conformal mapping equivalent to saying that the coordinate system is flat? [a]

Response. No. Flatness and conformality are different statements. A conformal map may use curved coordinate lines; what it preserves is the angle between intersecting curves. A Mercator-type map projection is a familiar example of local shape preservation with substantial scale distortion.

[a]Lecture timestamp: 00:55:35.

In two dimensions the required transformations are generated locally by analytic functions of a complex coordinate

$$z = \sigma + i\tau. \tag{1.15}$$

Finite boundaries can change appearance dramatically. A square region may be mapped to a smooth region, with the original corners sent to ordinary points on the new boundary rather than to its visually prominent points. There is no contradiction: conformality is a local interior condition, and the map preserves infinitesimal angles wherever it is regular.

Question from the audience. Is this coordinate freedom the reason the earlier join–split diagram could be chosen symmetrically? [a]

Response. Yes. The same conformal freedom that preserves the worldsheet equations lets us replace the awkward interaction strip by a more symmetric domain. The remaining freedom does not remove the physical join–split parameter; it repackages that parameter as the relative position of boundary insertions.

[a]Lecture timestamp: 00:56:32.

1.6 The Slit Sheet and the Disc

Return to four-string scattering. Two open strings enter, their endpoints join, the combined string propagates, and it later splits. Drawn as a flat worldsheet, the process is an infinite strip slit from each end. The separation τ_0 between the tips of the slits is the lifetime of the intermediate string, and the amplitude integrates over it.

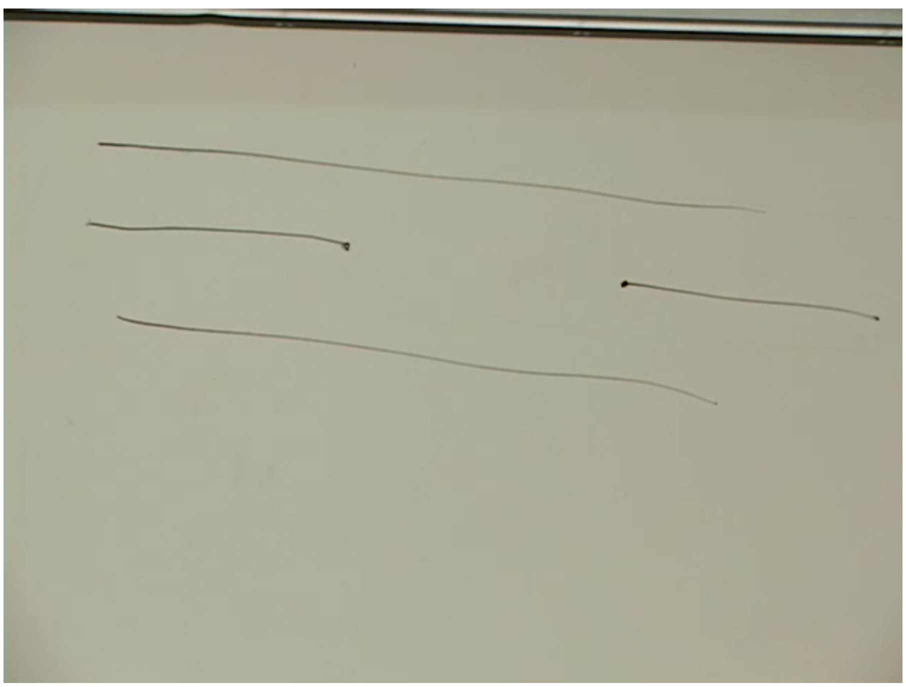

Figure 1.11: The doubly slit strip representing two open strings joining, propagating together, and splitting again.

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After Euclideanization, the embedding fields on this region satisfy the Laplace equation. Conformal invariance therefore permits the slit strip to be mapped to a disc. One way to see the topology is to foliate the disc by concentric circles and continuously deform those closed curves until they fill the slit region. The ends at infinity become exceptional boundary points under the map.

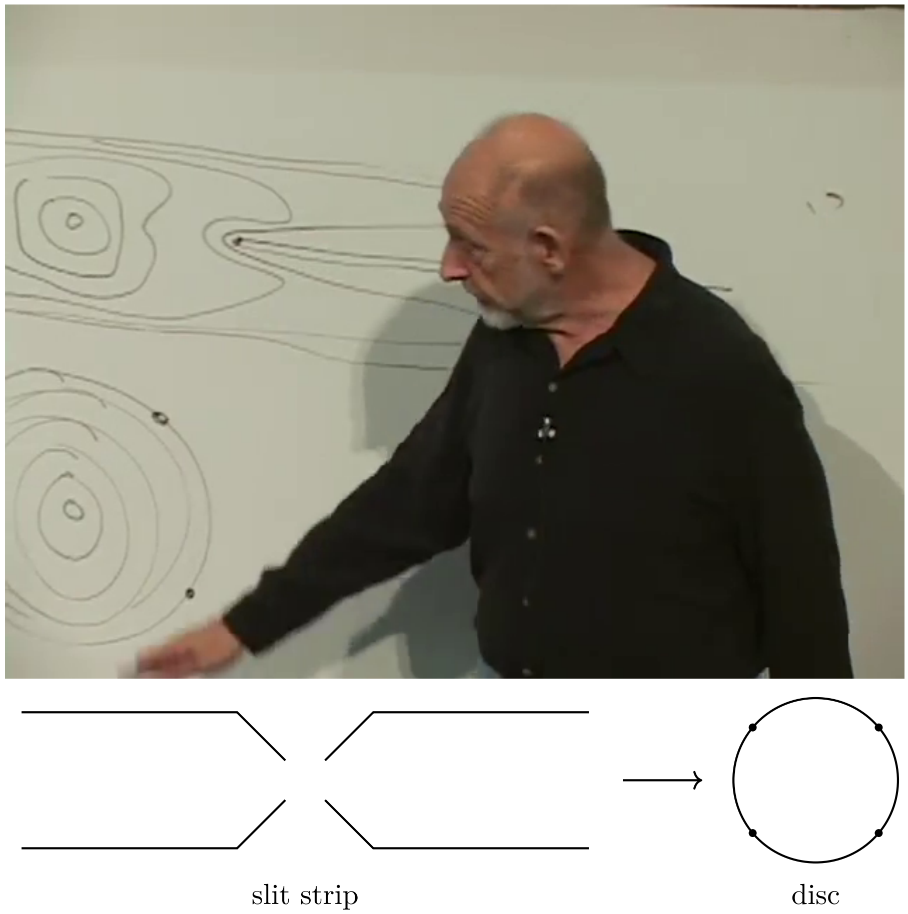

Figure 1.12: The board's level-curve comparison and a clean schematic of the conformal map from the slit strip to a disc. The four external ends become four special boundary points.

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The external states are not lost in this change of coordinates. Momentum, particle type, and oscillator level must be inserted at the four exceptional points. The formal objects carrying that information are vertex operators. Their detailed construction is deferred, but their role is already clear: they turn an otherwise homogeneous worldsheet path integral into a specified scattering experiment.

After using conformal transformations of the disc to move three boundary points to convenient reference positions, one real relative position remains. It may be represented by the cross-ratio of the four ordered boundary points. This is the disc version of the old lifetime τ_0.

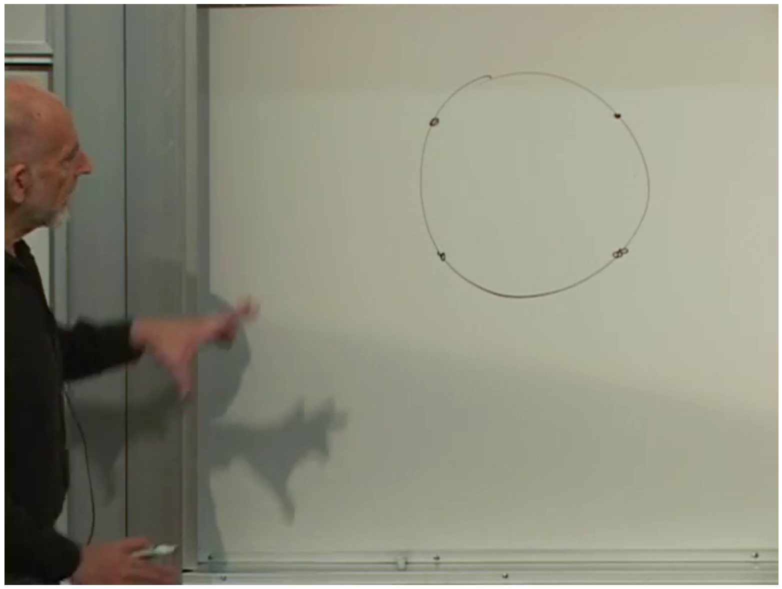

Figure 1.13: Four insertion points on the boundary of the disc. Conformal freedom removes all but one real modulus of their ordered configuration.

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The lifetime integral has therefore not disappeared. It has become an integral over that remaining modulus. With a conventional coordinate $0 < z < 1$, the four-point open-string integral takes the same form found in the preceding lecture,

$$\mathcal{A}(s,t) \propto \int_0^1 z^{-s-1}(1-z)^{-t-1}\, dz. \tag{1.16}$$

1.7 Two Limits of One Surface

At one end of moduli space, the two insertion points associated with the incoming pair approach one another. In the slit-strip coordinates this is a long interval between joining and splitting: a direct-channel intermediate string propagates for a long time. At the other end, a different neighboring pair approaches. The same surface is then naturally read as a crossed- channel exchange. These descriptions look unrelated on the strip but are neighboring boundary limits of one disc integral.

Returning to the mostly-plus convention of the previous lecture,

$$-s = (k_1 + k_2)^2, \qquad -t = (k_1 + k_3)^2. \tag{1.17}$$

The direct limit organizes poles in s; the crossed limit organizes poles in t. [3]

[3] Editorial clarification: Near the end of the lecture the board uses a shorthand sign for the Mandelstam invariants while also identifying s with positive center-of-mass energy squared. Equation (1.17) keeps the mostly-plus convention established in Lecture 6.

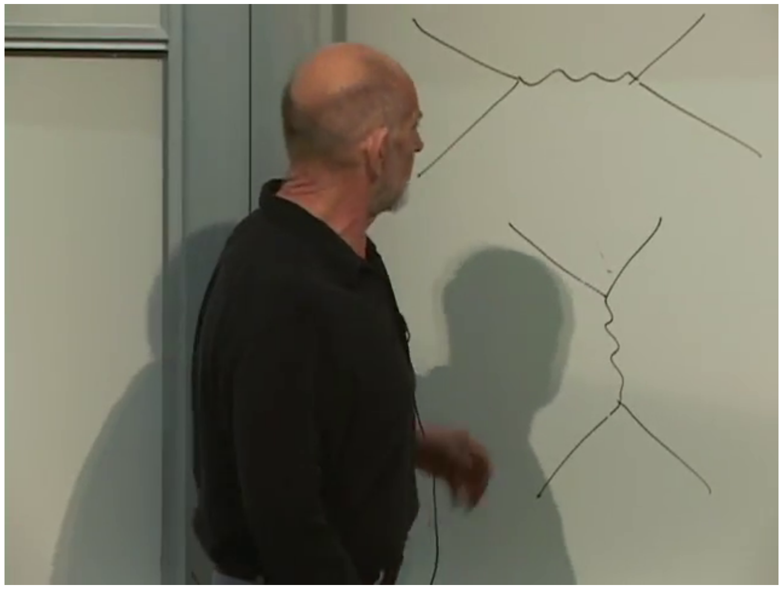

Figure 1.14: The two channel diagrams: long propagation in one orientation and exchange in the crossed orientation.

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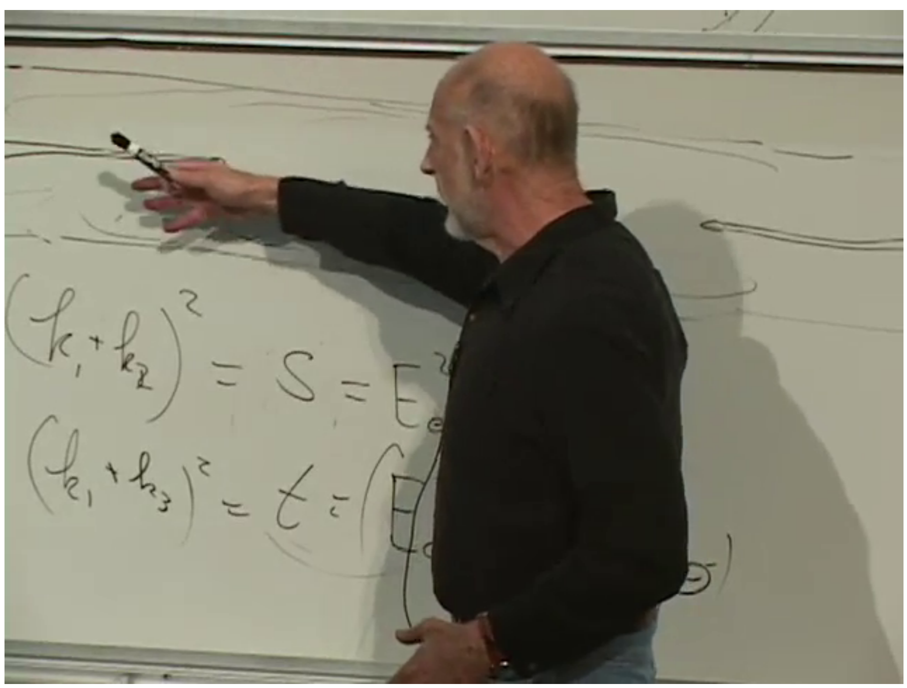

Figure 1.15: The board returns to the s- and t-channel momentum pairings while comparing the two limits of the disc modulus.

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One analytic object therefore supplies both expansions:

$$\mathcal{A}(s,t) = \mathcal{A}(t,s). \tag{1.18}$$

This is channel duality. In the original resonance language it appeared miraculous that a sum over direct-channel states could also encode crossed- channel exchange. On the Euclidean worldsheet, conformal symmetry explains why the two descriptions belong to the same surface. Historically this interpretation followed the 1969 discovery of the Veneziano amplitude very quickly.

Question from the audience. Is there a slit-strip picture corresponding to the crossed orientation as well? [a]

Response. Yes, but the direction called time must be chosen differently. The same Euclidean surface can be sliced so that another pair of external ends lies in the remote past and future. That alternative slicing is precisely why a single worldsheet admits more than one channel interpretation.

[a] Lecture timestamp: 01:20:49.

Question from the audience. Is the time for which the strings remain joined determined by the coupling constant? [a]

Response. No fixed lifetime is selected. The lifetime is an integration variable and every value contributes. The coupling controls the weight for joining and splitting—equivalently, the decay strength—and enters multiplicatively in the amplitude; it does not replace the modulus integral.

[a] Lecture timestamp: 01:21:39.